

- 18 A musical instrument is made using a long tube with a mouthpiece at one end. The other end is open and flared, as shown.



A musician maintains stationary sound waves with a node at the mouthpiece and an antinode at the other end. The fundamental frequency of sound that the instrument can produce is 80 Hz.

Which different frequencies of sound can be produced by the instrument?

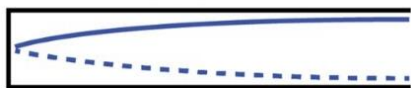
- A 160 Hz, 240 Hz, 320 Hz
- B 160 Hz, 320 Hz, 640 Hz
- C 240 Hz, 400 Hz, 720 Hz
- D 240 Hz, 320 Hz, 480 Hz

Ans: C

Since the situation is a one end open, one end closed pipe of length L ,

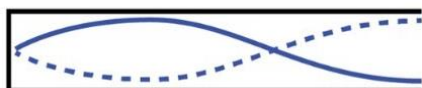
Fundamental harmonic: $\lambda = 4L \rightarrow f = 80\text{Hz}$

1/4 wave



1st Overtone: $\lambda = (4/3)L \rightarrow f = 3 \times 80\text{Hz} = 240\text{ Hz}$

3/4 waves



2nd Overtone: $\lambda = (4/5)L \rightarrow f = 5 \times 80\text{Hz} = 400\text{ Hz}$

5/4 waves



We can see that subsequent overtones are just odd-integer multiples of the fundamental frequency, in other words: $f, 3f, 5f, 7f, 9f, \dots, (2n+1)f$ are all possible solutions.

19 A portable fan battery is charged by connecting it to a constant potential difference of 6.0 V . The